SPOTIFY 2023
DATA ANALYSIS

Power BI Report

Data Cleaning • DAX Measures • Visual Analysis

Prepared by	Maninder Singh
UID	24BAI70399
Institution	Chandigarh University
Subject	Business Intelligence
Tool	Microsoft Power BI Desktop (Power Query Editor)
Dataset	spotify-2023.csv
Date	28 September 2026

Enhanced edition with original Power BI screenshots

1. Objective & Dataset Overview

The objective is to import the Spotify 2023 songs dataset into Power BI, clean and prepare it using Power Query, create calculated measures with DAX, and build interactive visuals to explore tracks, artists, audio features and performance across Spotify, Apple and Deezer.

Dataset size: 953 rows and 24 columns before cleaning; 943 rows after duplicate and blank-row cleaning.

Key fields include track and artist identifiers, release date fields, Spotify streams, platform chart/playlist fields, BPM, key, mode and several audio characteristics.

2. Data Import

The CSV was loaded through Get Data. The original workflow selected Western European (Windows) encoding, comma delimiter and Transform Data before loading the prepared table.

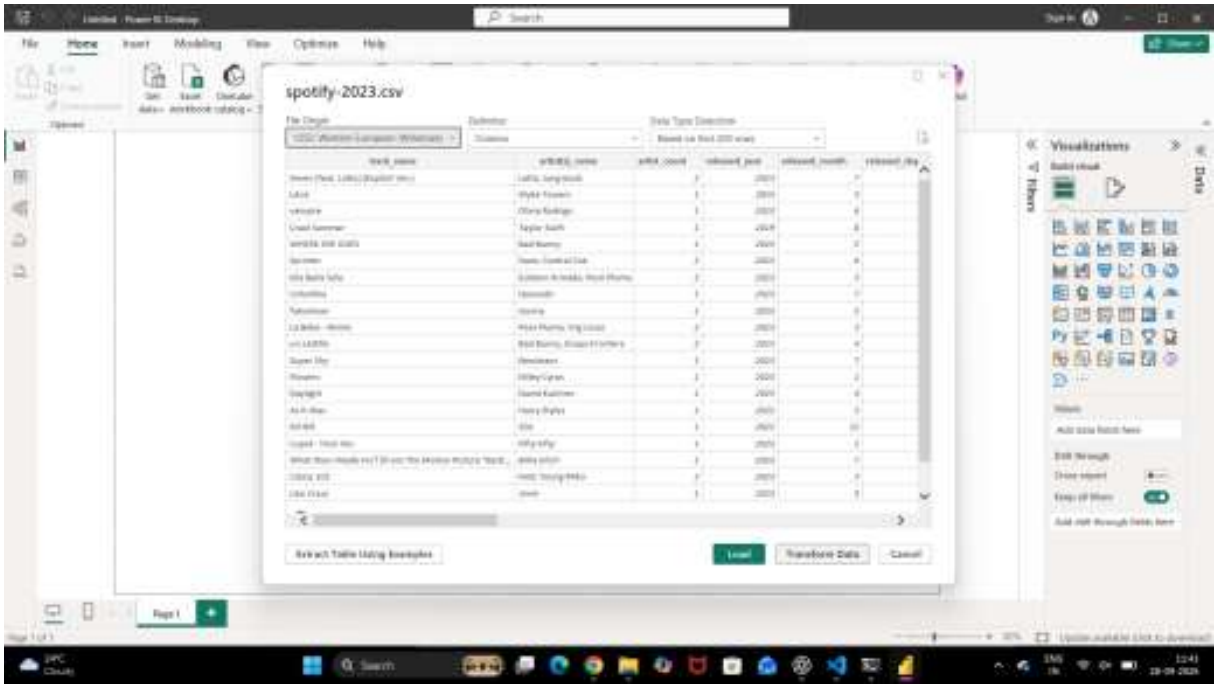


Figure 1. Power BI import dialog for spotify-2023.csv.

3. Power Query Cleaning

The cleaning sequence documented in the report is: Promoted Headers → Changed Type → Removed Duplicates → Cleaned Text → Removed Blank Rows.

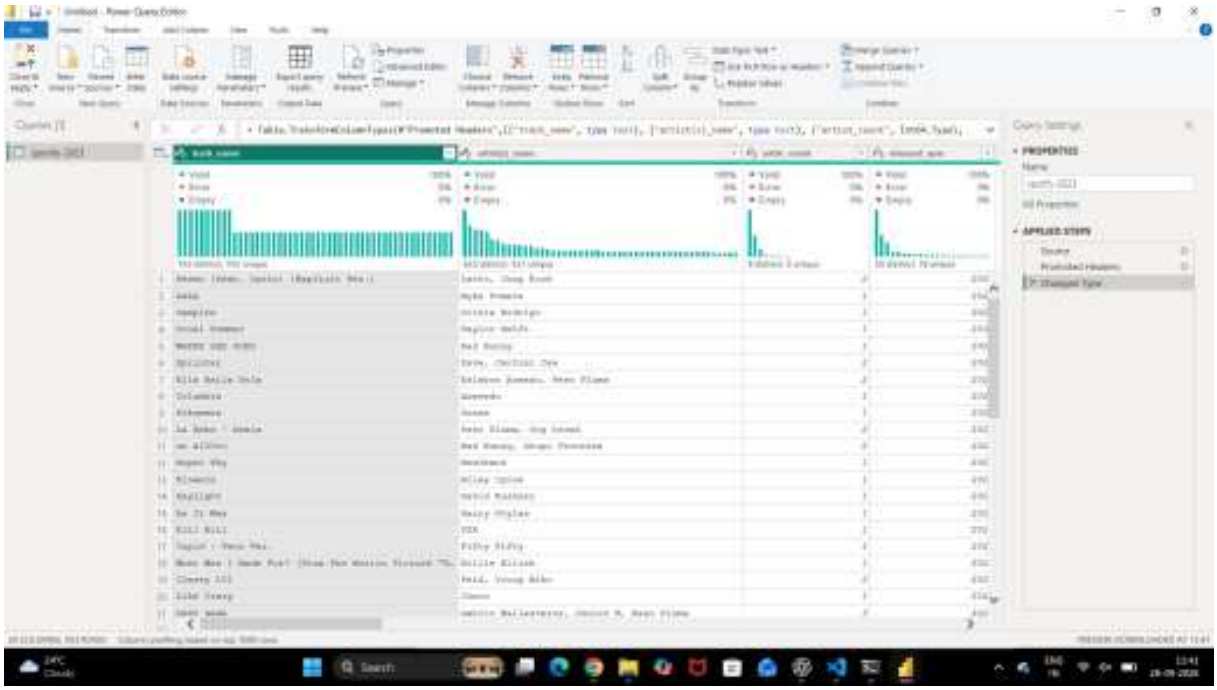


Figure 2. Power Query Editor with promoted headers and data types.

3. Power Query Cleaning — Screenshots

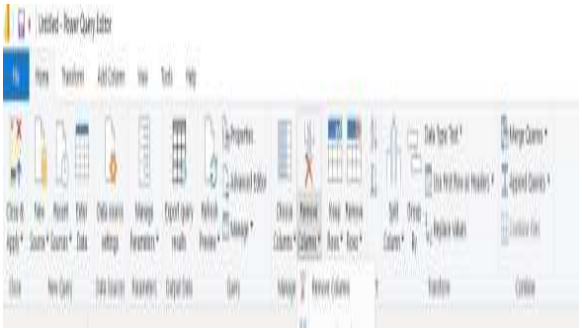


Figure 3. Power Query ribbon / column operations.

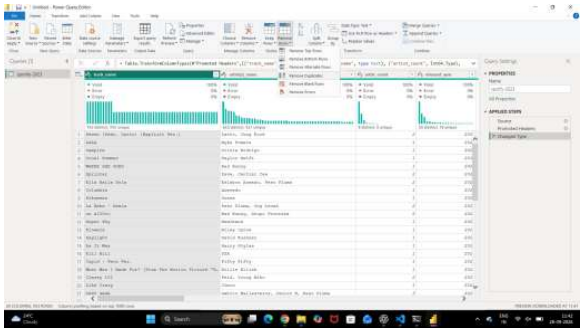


Figure 4. Power Query table after transformation.



Figure 5. Applied cleaning steps.

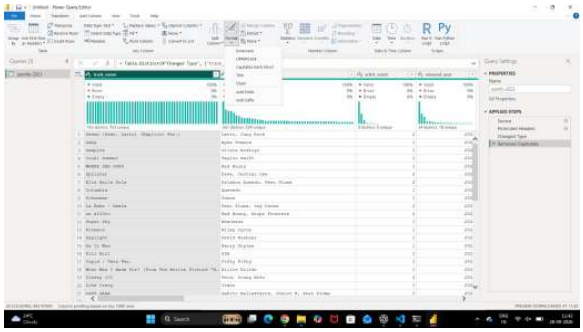


Figure 6. Cleaned Power Query view.

Cleaning result: duplicate removal reduced the dataset from 953 rows to 943 rows. Text cleaning removes non-printable characters and blank-row removal eliminates empty records.

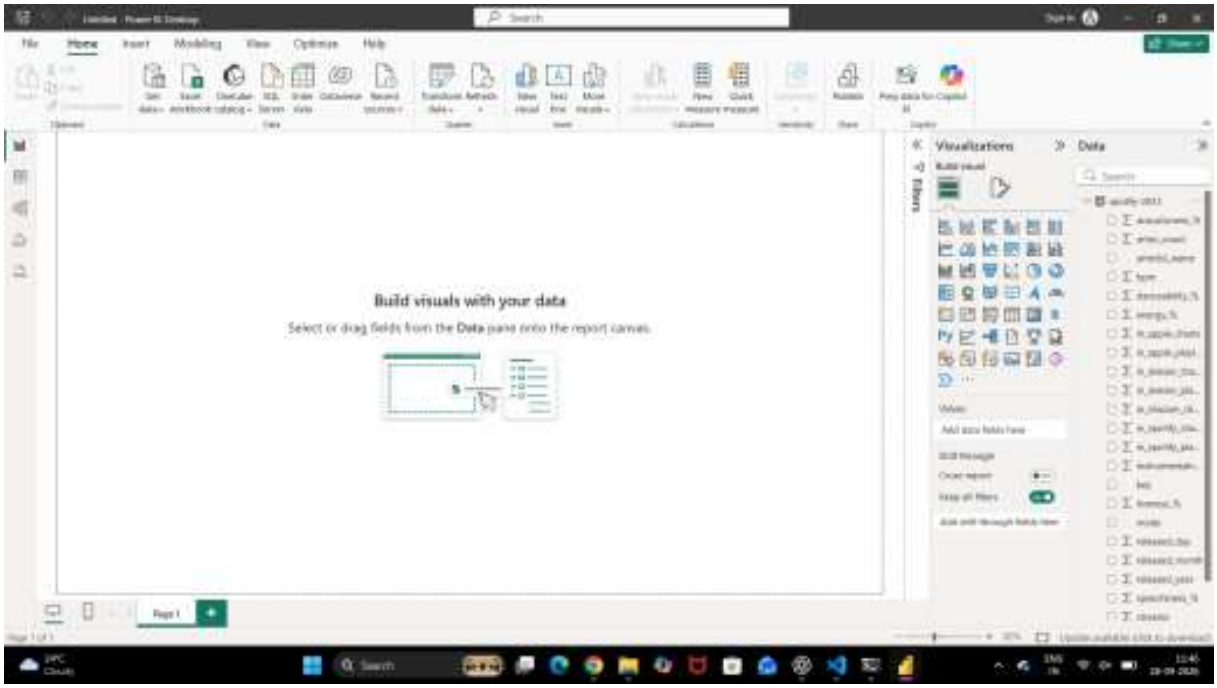


Figure 7. Power BI report canvas after loading the prepared data.

4. DAX Measures

Measures were created using Modeling → New measure. The original report uses AVERAGE-based measures for streams, energy, Apple playlists, Deezer playlists, Spotify playlists and danceability, plus Total Streams for visualizations.

Measure	Formula / Use
Average Streams	AVERAGE('spotify-2023'[streams])
Average Energy	AVERAGE('spotify-2023'[energy_%])
Avg Apple Playlists	AVERAGE('spotify-2023'[in_apple_playlists])
Avg deezer Playlists	AVERAGE('spotify-2023'[in_deezer_playlists])
Avg Spotify Playlists	AVERAGE('spotify-2023'[in_spotify_playlists])
Average Danceability	AVERAGE('spotify-2023'[danceability_%])
Total Streams	Used on the axes of the stream visuals

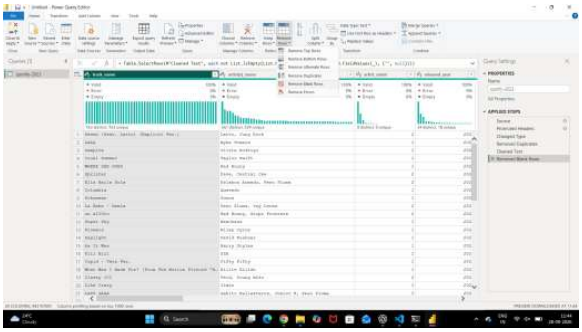


Figure 8. Report fields after cleaning.

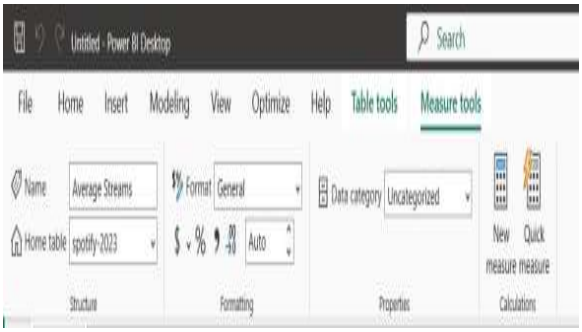


Figure 9. Naming / formatting a DAX measure.

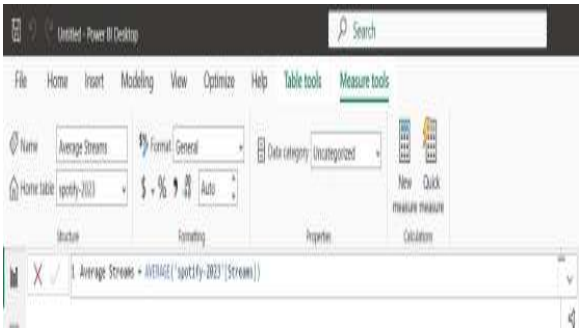


Figure 10. Average Streams formula.

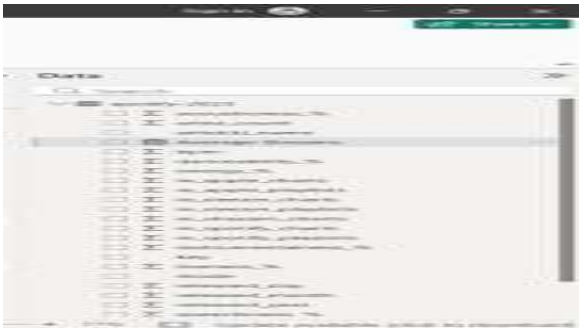


Figure 11. Measure visible in the Data pane.

4. DAX Measures — Additional Screenshots

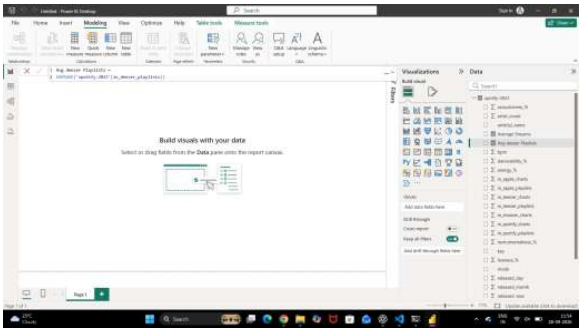


Figure 12. Measure creation / formula view.

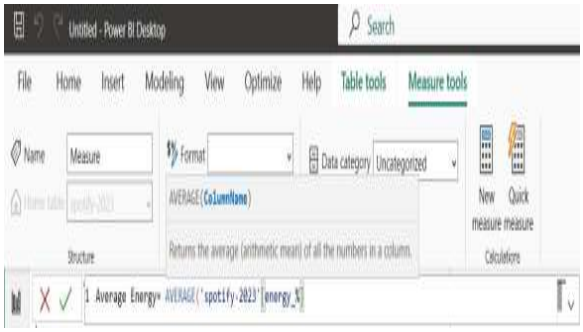


Figure 13. Average Streams measure formula.

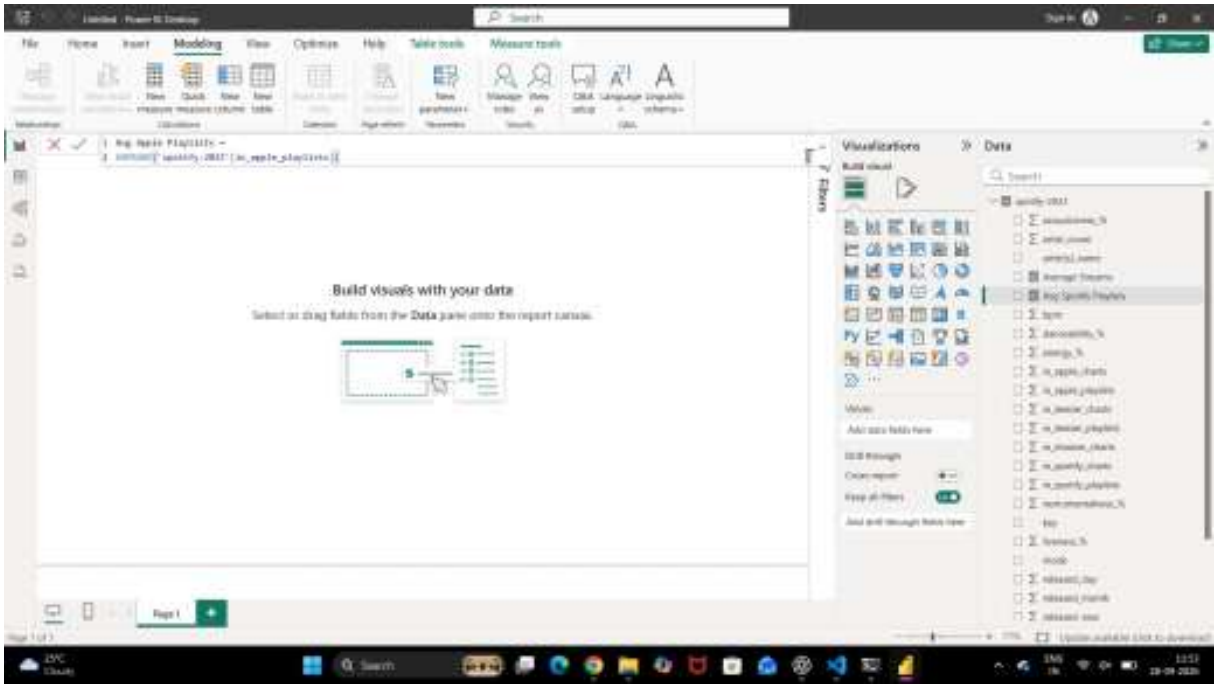


Figure 14. Average Energy measure.

These screenshots document the measure-building workflow and support reproducibility of the Power BI report.

5. Report Page 1 — Top Tracks by Streams

A clustered bar chart plots Total Streams against track_name. The original analysis identifies Blinding Lights and Shape of You as the leading tracks at roughly 3.5–4 billion streams each, followed by Someone You Loved, Dance Monkey and Sunflower (Spider-Man: Into the Spider-Verse).

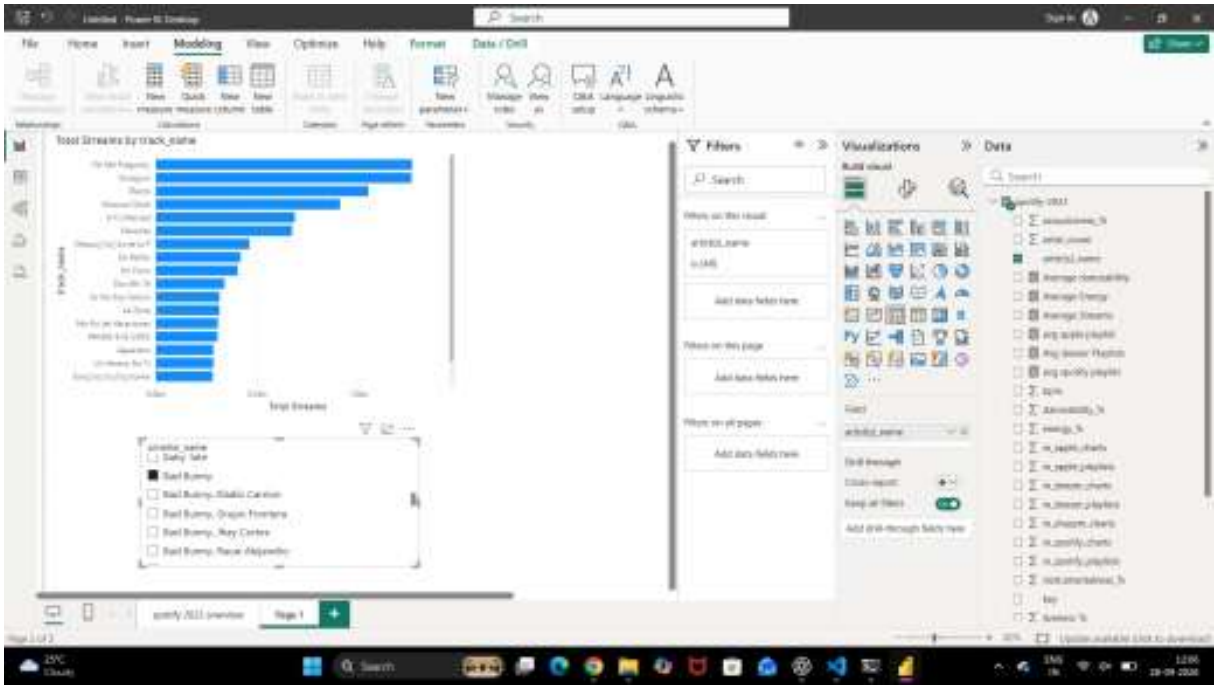


Figure 15. Total Streams by track_name.



Figure 16. Track chart filtered using the artist slicer.

6. Report Page 2 — Audio Features vs Streams

Bubble/scatter charts compare Total Streams with danceability, energy and liveness. artist(s)_name is used in the Legend and Sum of streams controls bubble size.

The analysis shows a small number of artists far above the rest at roughly 13–14 billion total streams, while most artists cluster near the bottom-left.

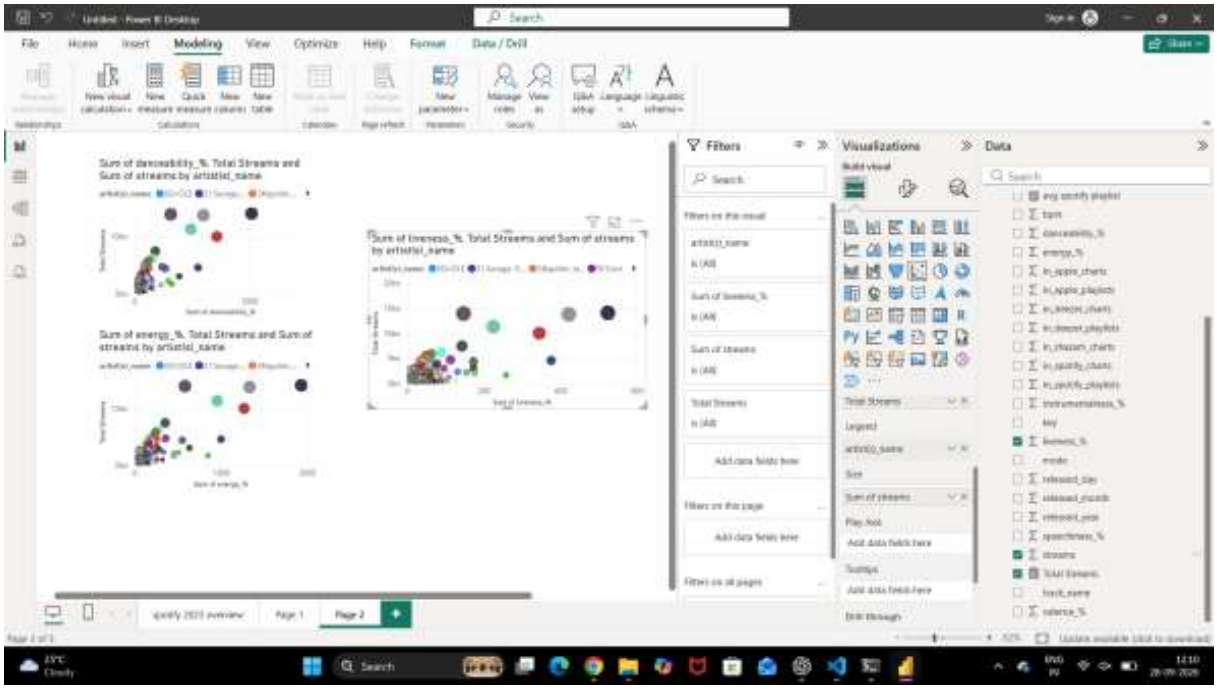


Figure 17. Danceability vs Total Streams by artist.

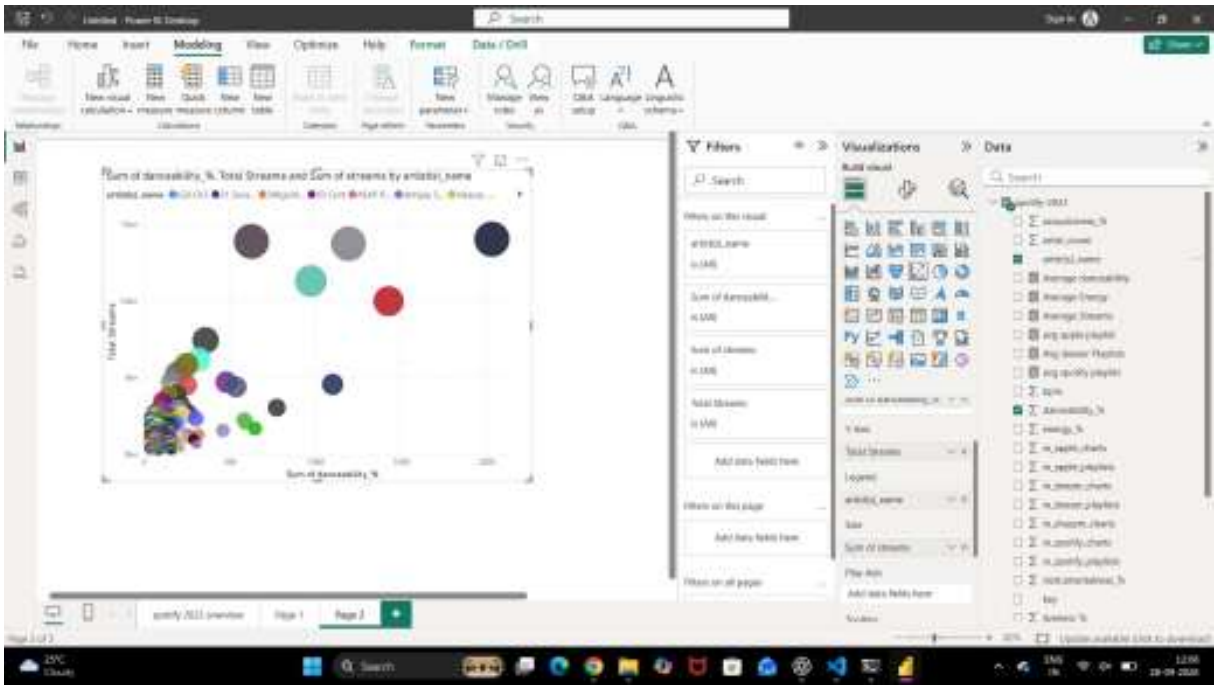


Figure 18. Energy / liveness bubble-chart view.

Important interpretation note: the X-axis uses Sum of the feature percentage. Artists with more songs can therefore score higher simply because more values are added. Using Average Danceability or Average Energy gives a fairer artist-level comparison.

7. Report Page 3 — Platform Presence by Artist

A multi-row card displays sums of Apple charts/playlists, Deezer charts/playlists and Spotify charts/playlists by artist.

For example, the report shows (G)I-DLE with 881 Spotify playlist placements, 242 Apple charts and 24 Apple playlist entries. 24kgoldn, Iann Dior is shown with about 13K Spotify playlist placements.

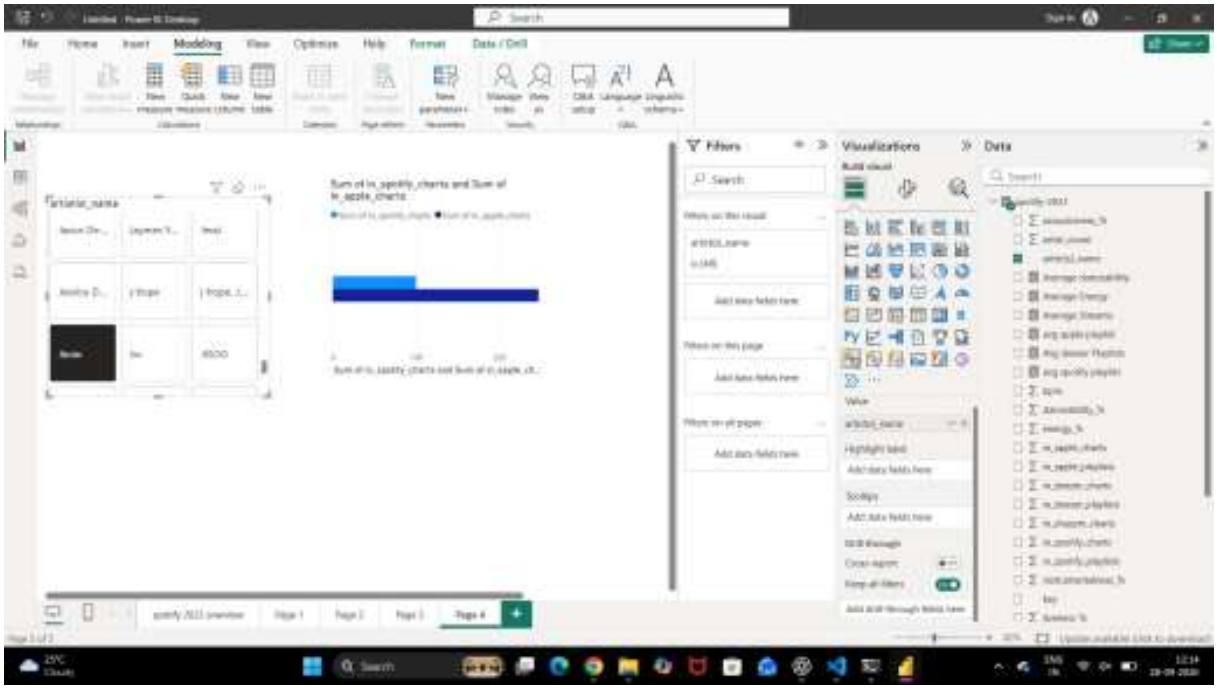


Figure 19. Platform chart and playlist counts by artist.

8. Report Page 4 — Spotify vs Apple Charts

A tile-style artist slicer controls a bar chart comparing Sum of Spotify chart appearances with Sum of Apple chart appearances.

With Jimin selected, Apple is shown at about 250 versus roughly 100 on Spotify. With Coldplay selected, Spotify is about 70 versus roughly 25 on Apple.

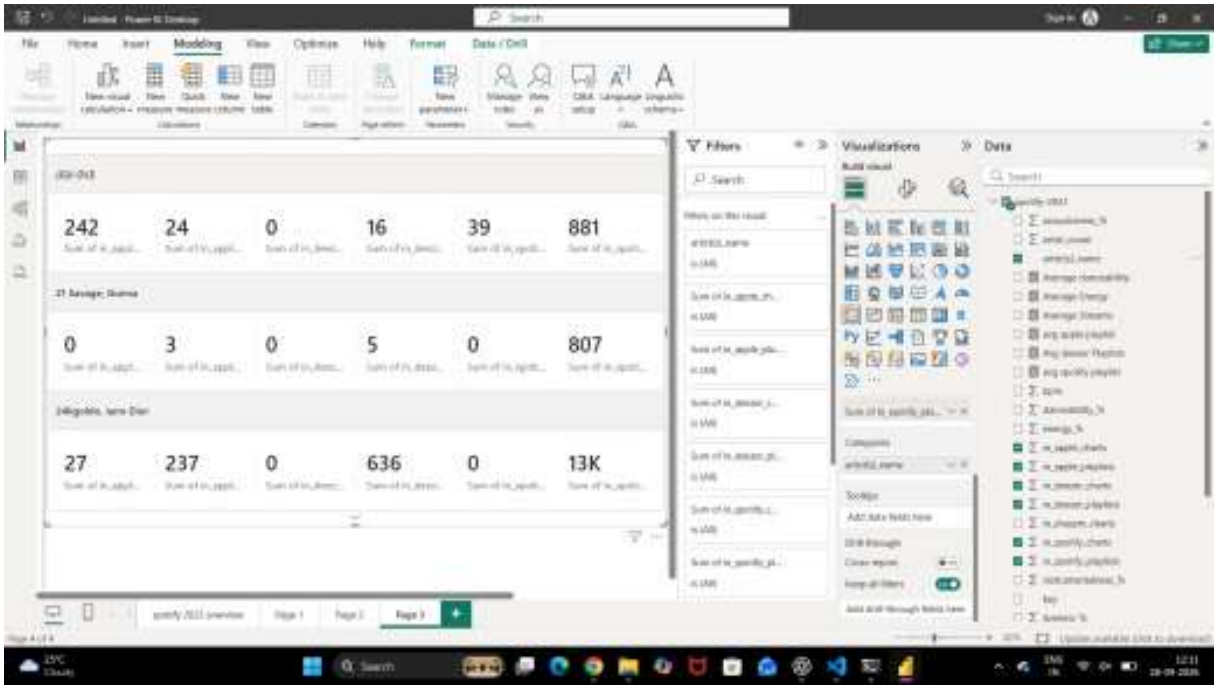


Figure 20. Jimin selected — Spotify vs Apple chart comparison.

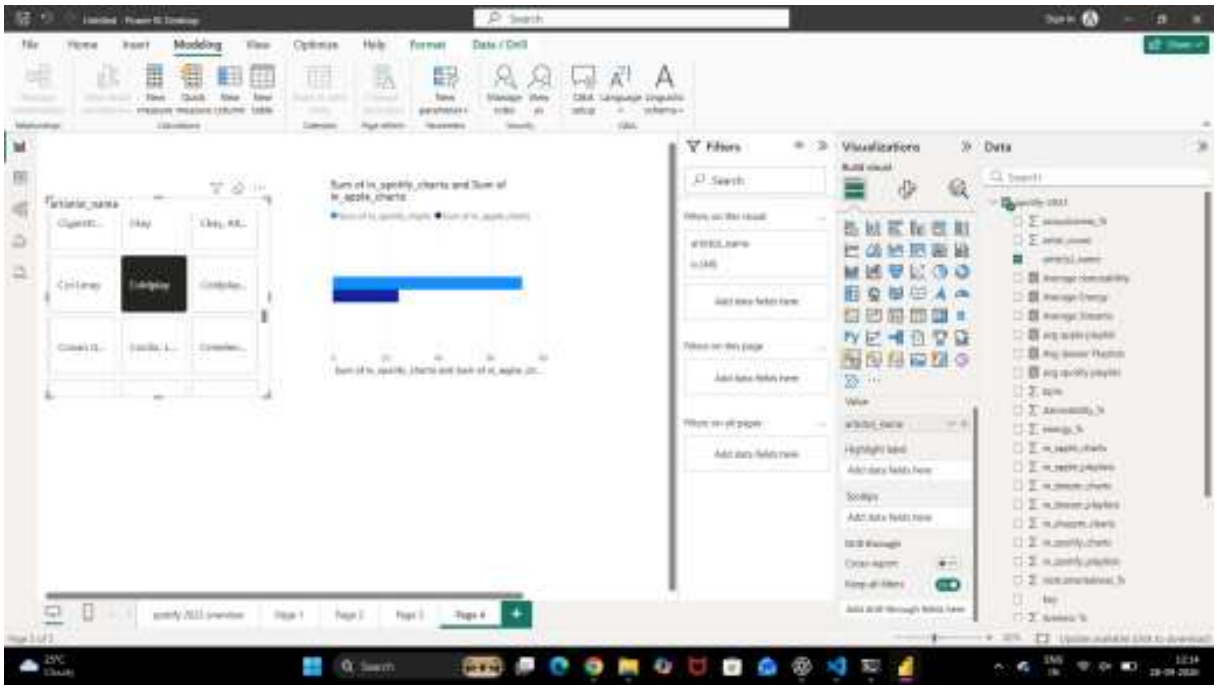


Figure 21. Coldplay selected — Spotify vs Apple chart comparison.

9. Report Page 5 — Release Month Filter

A numeric range slicer on released_month is paired with a track_name list. Moving the slider narrows the displayed songs to the selected release-month range.

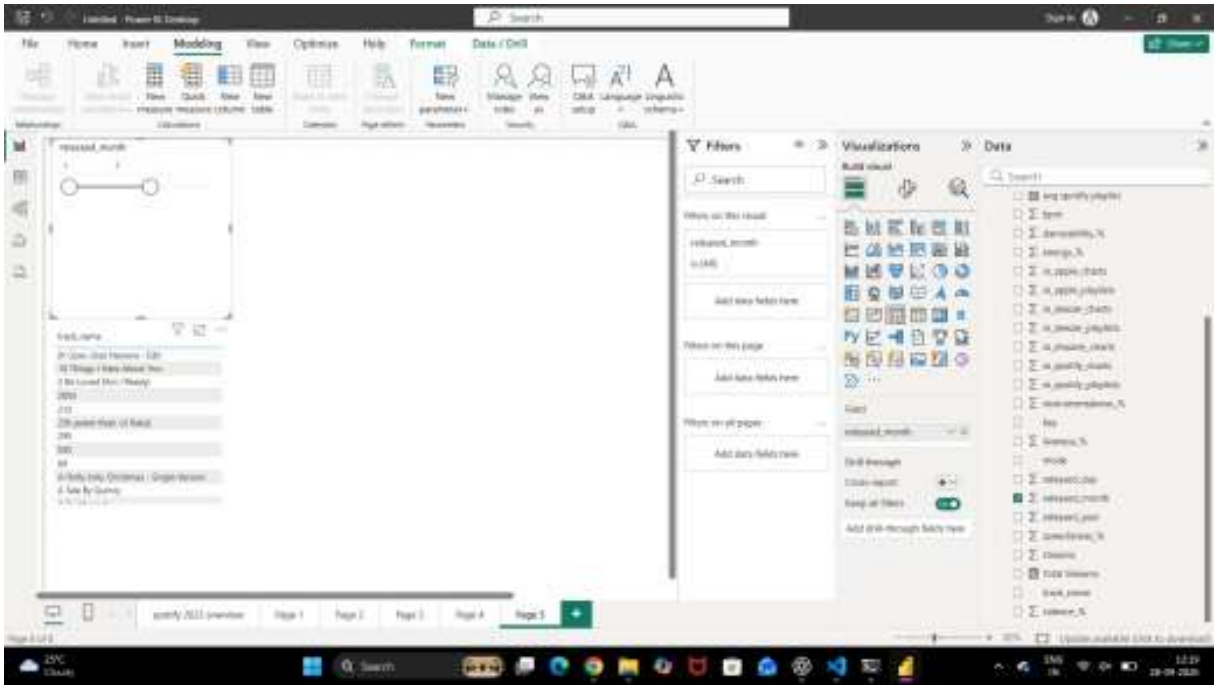


Figure 22. Release-month filter and track list.

10. Key Findings

1. A few tracks, led by Blinding Lights and Shape of You, have streaming totals far above the rest of the dataset.

2. Total streams are highly concentrated: a handful of artists reach roughly 13–14 billion streams while most sit at a small fraction of that level.

3. Platform performance varies by artist; the report shows Jimin with stronger Apple chart presence in the selected visual and Coldplay with stronger Spotify chart presence.

4. Spotify playlist reach is substantially larger than Apple or Deezer for the artists shown on the platform card; 24kgoldn, Iann Dior is given as an example.

11. Conclusion

Power Query was used to remove duplicates, clean text and blank rows, reducing the dataset to 943 records. DAX measures provide reusable averages, while five report pages use bar charts, bubble charts, cards and slicers to make the analysis interactive.

The report highlights concentrated streaming success among a small set of tracks and artists and shows differences in chart performance between Spotify and Apple. The included screenshots document the Power BI workflow and make the report easier to verify and present.

End of Report